RAJALAKSHMI ENGINEERING COLLEGE

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C2

Started on	Thursday, 17 September 2026, 8:09 AM
State	Finished
Completed on	Thursday, 17 September 2026, 8:18 AM
Time taken	9 mins 6 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    int count = 0;

    // Read the positive integer n
    if (scanf("%d", &n) != 1) {
        return 0;
    }

    count++; // 1 for the if(n==1) condition check
    if (n == 1) {
        count++; // 1 for the printf("*") operation inside the block
    }
    else {
        count++; // 1 for the outer loop initialization (int i = 1)

        for (int i = 1; i <= n; i++) {
            count++; // 1 for outer loop condition check (i <= n)
            count++; // 1 for inner loop initialization (int j = 1)

            for (int j = 1; j <= n; j++) {
                count++; // 1 for inner loop condition check (j <= n)
                count++; // 1 for the print block execution
                break;
            }

```

```
            count++; // 1 for outer loop increment (i++)
        }
    }

    // Print the final count value as required
    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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